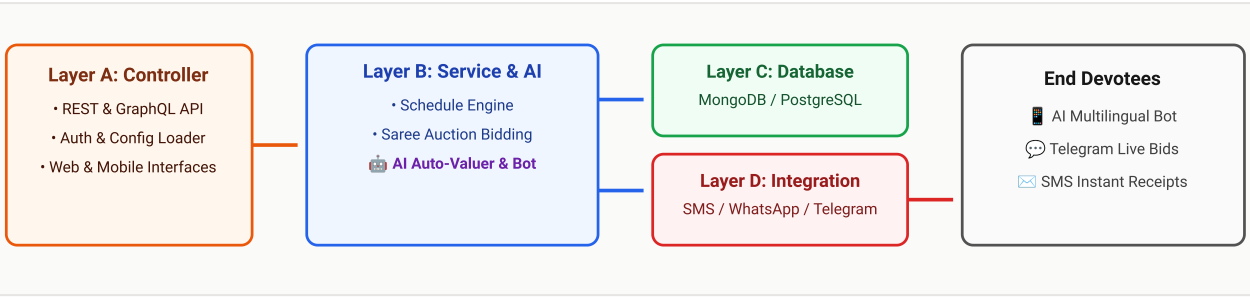
Temple Management System (TMS)

AI-Powered, Open-Source (GitHub Modular Template) & Multi-Channel System Specification

DISTRIBUTION MODEL	CORE TECH	DATABASE	AI INTEGRATION
GitHub Open Repository	Node.js / Python AI Engine	MongoDB / PostgreSQL	LLM Chatbots & Vision APIs

1. SYSTEM ARCHITECTURE OVERVIEW (LAYER BREAKDOWN)

The Temple Management System is designed as an **open-source, template-driven software engine**. Any temple committee can clone the repository from GitHub and customize settings via a simple configuration file (`temple.config.json`).



2. CORE FUNCTIONAL PILLARS

TIMING & SCHEDULE
Arti & Visarjan
AI dynamic crowd prediction & automated timing updates.

PRASAD MANAGEMENT
Mahaprasad
Token distribution, seating capacity & slot tracking.

FINANCIAL RECORDS
Collections
Online/offline donations with automated PDF receipts.

OFFERING AUCTION
Saree / Gold Bids
AI Image auto-tagging & live bid broadcast engine.

3. GITHUB REPOSITORY STRUCTURE & SELF-HOSTING STRATEGY

The project will be released as an open-source GitHub repository (e.g., `github.com/open-temple/temple-management-system`). Any temple admin can clone it and customize local temple parameters, branding, payment gateways, and notification keys via simple environment variables.

```
// Example GitHub Project File Structure temple-management-system/ ├── .env.example # Database, WhatsApp, AI API keys setup ├── temple.config.json # Self-configuration (Temple Name, Timings, Currency) ├── docker-compose.yml # One-click deployment (Node.js + MongoDB + Redis) ├── src/ │ ├── controllers/ # Layer A: REST API endpoints & Webhooks │ ├── services/ # Layer B: Arti Scheduler, Auction Bidding, Prasad │ ├── ai_engine/ # AI Agents: Multilingual Bot & Image Auto-Tagger │ └── notifications/ # Layer D: WhatsApp, Telegram & SMS Adapters └── admin-portal/ # Responsive Web Dashboard (React/Next.js)
```

4. TEMPLE CUSTOMIZATION FILE (TEMPLE.CONFIG.JSON)

Instead of hardcoding temple data, each temple defines its unique attributes in a simple JSON file without modifying backend code:

```
{ "templeName": "Shree Siddhivinayak Temple Trust", "location": "Mumbai, Maharashtra, India", "supportedLanguages": ["en", "hi", "mr", "gu"], "currency": "INR", "modulesEnabled": { "artiTimings": true, "visarjanAlerts": true, "mahaprasadTokens": true, "sareeAuction": true }, "aiEngine": { "provider": "openai", // or "gemini" / "ollama_local" "multilingualBotEnabled": true, "autoValueAuctionItems": true } }
```

5. EASY DEPLOYMENT WORKFLOWS FOR ANY TEMPLE

Deployment Options for Temple Administrators:

- **Option A (1-Click Cloud - Recommended):** Deploy directly to free or low-cost cloud tiers (Render, Railway, AWS Lightsail) using the repository's `Dockerfile`.
- **Option B (Local Temple Server):** Run locally on a temple office computer using Docker Desktop without requiring expensive internet bandwidth.

6. MANDATORY AI ENGINE INTEGRATION FRAMEWORK

AI is embedded directly into Layer B to automate labor-intensive temple management tasks, streamline devotee communication, and enhance live auctions.

🤖 **AI-Powered Capabilities Overview:**

- **1. Multilingual Devotee Assistant:** AI WhatsApp/Telegram bot capable of instantly answering queries regarding Arti, Visarjan, and Mahaprasad timings in Marathi, Hindi, Gujarati, English, and local dialects.
- **2. Vision AI for Saree & Offering Cataloging:** Automatically analyzes uploaded photos of gifted sarees/clothes, identifies fabric type (e.g., Silk, Cotton), detects color, and suggests a recommended base auction price.
- **3. Smart Crowd & Prasad Queue Optimization:** Predicts festival rush times using historical attendance data to dynamically stagger Mahaprasad slot distribution.

AI Feature	Technical Implementation	Value to Temple Administration
Multilingual AI Bot	LLM API (OpenAI/Gemini/Local Llama) integrated with WhatsApp Meta API & Telegram Webhooks.	Handles thousands of repetitive devotee questions daily without requiring dedicated temple call center staff.
Saree / Gift Vision Tagger	Computer Vision AI model auto-classifies uploaded item images into categories (Gold, Saree, Shawl, Silver).	Accelerates auction item entry by 80%; admin just snaps a picture and AI generates title & description.
Fair-Bid AI Sentinel	Anomaly detection algorithms monitoring live bidding stream for fraudulent or fake bids.	Ensures high integrity during sacred saree and ornament auctions by flagging suspicious accounts.
Auto-Receipt OCR	OCR engine extracts payment details from manual cheque or bank upload slips for instant ledger entry.	Reduces accountant data entry errors during high-volume donation seasons.

7. MULTI-CHANNEL NOTIFICATION ROUTER (LAYER D)

The messaging pipeline routes announcements and transaction receipts through a fallback strategy (WhatsApp → Telegram → SMS) to guarantee high delivery at minimal cost.

Channel	Primary Purpose	Cost Profile	Fallback Priority
WhatsApp	Interactive AI Chatbot, Auction Bidding, Payment Receipts	Low (~$0.005/msg)	Priority 1 (Primary)
Telegram	Free Broadcast Channel for Arti timings & Live Auctions	100% Free	Priority 2 (Broadcast)
SMS	Urgent Payment Verification & Offline Devotee Alerts	Moderate (~$0.002/msg)	Priority 3 (Fallback)

8. DATABASE SCHEMAS WITH AI METADATA

MongoDB dynamic document schemas with embedded AI metadata fields for intelligent cataloging.

Collection / Table	Key Attributes / Fields	Primary Index & Constraints
`schedules`	`event_type` (Arti/Visarjan/Puja), `start_time`, `end_time`, `ai_predicted_crowd_level`	`Index: {start_time: 1, event_type: 1}`
`collections`	`transaction_id`, `devotee_id`, `amount`, `payment_mode`, `purpose`, `receipt_url`	`Unique: transaction_id`
`auctions`	`item_id`, `title`, `ai_generated_tags`, `base_price`, `current_bid`, `status`	`Index: {status: 1, end_time: 1}`
`bids`	`auction_id`, `devotee_phone`, `bid_amount`, `bid_timestamp`, `ai_risk_score`	`Index: {auction_id: 1, bid_amount: -1}`

9. OFFERING & SAREE AUCTION LIFECYCLE WITH AI WORKFLOW

End-to-End AI-Assisted Auction Process:

1. **Photo Capture & AI Analysis:** Admin takes a picture of the donated Saree/Cloth. AI automatically tags material (e.g., Paithani Silk) and suggests a starting bid.
2. **Broadcasting:** Scheduled auction catalog is automatically formatted and published to Telegram Channels and WhatsApp Subscriber Lists.
3. **AI-Monitored Live Bidding:** Devotees place bids directly through WhatsApp reply commands or Telegram Bot. AI flags anomalous bids.
4. **Automated Settlement:** The winning bidder receives an automated checkout link via SMS/WhatsApp with digital receipt delivery upon completion.

10. SECURITY & DATA GOVERNANCE

- **Data Privacy & Local Storage:** Devotee phone numbers and financial data remain on the temple's private cloud/database.
- **Configurable API Guardrails:** Rate-limiting on AI endpoints prevents API quota overuses during major festival spikes.
- **Audit Logging:** Immutable logs for all donation receipts and auction winners to maintain total transparency for temple trustees.

11. ESTIMATED DEVELOPMENT EFFORT & WORK BREAKDOWN (WBS)

Total estimated effort to build the initial open-source GitHub repository with AI capabilities is **9 to 11 Weeks**.

Phase	Key Deliverables	Est. Hours	Timeline
Phase 1: Open GitHub Template Core	Configurable architecture (`temple.config.json`), Docker setup, Auth, DB models.	90 hrs	Weeks 1–2
Phase 2: Core Temple Services	Schedule engine (Arti/Visarjan), Mahaprasad queue, Donation collection ledger.	110 hrs	Weeks 3–4
Phase 3: AI Engine Integration	Multilingual Bot, Saree Vision Auto-Tagger, and Fair-Bid Sentinel algorithms.	110 hrs	Weeks 5–6
Phase 4: Saree & Gift Auction Engine	Bidding workflow, live catalog broadcasting via Telegram & WhatsApp.	90 hrs	Weeks 7–8
Phase 5: Web Admin Dashboard	Responsive mobile/desktop PWA web portal for counter staff and temple trustees.	80 hrs	Weeks 9–10
Phase 6: QA, Open Source Docs & Release	GitHub README setup, setup guide videos, security audits, and initial v1.0 release.	40 hrs	Week 11
TOTAL EFFORT	**Production-Ready Open Source Release**	**520 hrs**	**11 Weeks**

12. ESTIMATED COST & INFRASTRUCTURE BUDGET

Cost Component	Details / Specs	Est. Cost (USD / INR)
Software Development Effort	520 hours full-stack development, AI integration, and open-source documentation	$8,500 – $13,000 / ₹7L – ₹11L
Cloud Hosting / Server	Self-hosted Docker on cheap VPS (DigitalOcean / AWS) or local computer	$0 – $40 / month
AI API Usage (OpenAI / Gemini)	Pay-as-you-go based on AI conversation volume & image processing	~$10 – $30 / month
Telegram Bot API	Broadcast channels & bidding bot integration	**100% Free**

13. GITHUB COMMUNITY DEPLOYMENT CHECKLIST

1. `git clone https://github.com/open-temple/temple-management-system.git`
2. Copy `.env.example` to `.env` and fill in WhatsApp, Telegram, and AI API keys.
3. Modify `temple.config.json` with your temple name, logo URL, and active modules.
4. Run `docker-compose up -d` to launch the web dashboard and notification backend!